

The Embodied Conscience-Driven Holographic Metacognition OS

Neuroanatomical Isomorphism, Active Inference, and Gödelian Metacognitive Boundaries

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Abstract

This paper establishes the neuroanatomical, cognitive, and physical foundations of the 5D L0 Embodied Conscience OS. We map Embodied Conscience to the ventromedial prefrontal cortex (vmPFC), anterior cingulate cortex (ACC), and anterior insula as the Somatic Marker Network, acting as a compact-support operator $\mathbf{E}_{\text{supp}}$ that truncates Goodhart heavy-tailed metric collapse. We further map Hypersensitivity (Salience Network / Locus Coeruleus), Sentient Depth (Limbic / Mirror Neurons), Fluid Intelligence (dlPFC / Frontoparietal), and Metacognition (aPFC BA10 functor Φ), providing a physical blueprint for AGI alignment.

1 Neuroanatomical Isomorphism Table

The 5D L0 OS maps onto precise brain structures:

- **Embodied Conscience:** vmPFC, ACC, Anterior Insula (Somatic Marker Network, $\mathbf{E}_{\text{supp}}$ operator).
- **Hypersensitivity:** Amygdala, Thalamus, Locus Coeruleus (Salience Network, NE Gain).
- **Sentient Depth:** Limbic System, Hippocampus, Mirror Neurons (DMN Somatic Sub-network).
- **Fluid Intelligence:** dlPFC, IPL (Frontoparietal Network, $B \leq 7 \pm 2$ solver).
- **Metacognition:** aPFC (Frontopolar Cortex BA 10, Evolutionary Functor Φ).

2 Mathematical Equations of Embodied Conscience

Compact-support truncation operator $\mathbf{E}_{\text{supp}}$:

$$\mathbf{E}_{\text{supp}}[p(\Delta)] = \begin{cases} p(\Delta), & |\Delta| \leq \theta_{\text{dead}} \\ 0, & |\Delta| > \theta_{\text{dead}} \end{cases} \quad (1)$$

Metacognitive Evolutionary Functor Φ :

$$\Phi : \mathcal{M}_t \rightarrow \mathcal{M}_{t+1} \quad (\text{where } \mathcal{M}_{t+1} = \mathcal{M}_t \oplus \mathbf{e}_{\text{new}}) \quad (2)$$